Seat No._____

GANPAT UNIVERSITY
M.Sc. (CA & IT) SEM-II EXAMINATION
April - June 2023
P22A1ND: NOSQL DATABASE

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

SECTION-I

1. Answer the following: (Any Five) **(30)**

1. Define NoSQL database. Why is it so popular?
2. List out various types of NoSQL databases. Explain any 2 of them in brief.
3. How can you use various logical and comparison operators in MongoDB?
4. Explain the concept of Data Modeling in MongoDB? How can you manage embedded documents in MongoDB?
5. How can you update the record in MongoDB? Explain different field operators available in update method.
6. Explain steps to connect Node.js with MongoDB. Perform basic operations.
7. What are the different ways to take backup and restore in MongoDB?

SECTION-II

2. Answer the following: (Any Two) **(12)**

1. Explain CAP theorem in the context of NoSQL databases.
2. What are the key features of MongoDB? How can you create database and collections into it? Explain with proper example.
3. Explain various datatypes supported in MongoDB with relevant example.

3. Answer the following: (Any Two) **(12)**

1. What is indexing? How can you manage indexing in MongoDB?
2. Explain the concept of aggregation in MongoDB.
3. Differentiate between Redis Vs MongoDB NoSQL databases.

4. Answer the following: (Any One) **(06)**

1. Write a python script to perform insert, delete and find operations from MongoDB collection.
2. Explain steps to connect PHP with MongoDB. Perform basic CRUD operations

END OF PAPER

Seat No._____

GANPAT UNIVERSITY
M.Sc. (CA & IT) SEM-II EXAMINATION
April - June 2023
P22A2PYT: PYTHON PROGRAMMING

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
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3. Be precise and to the point in your answer

SECTION-I

1. **Answer the following:** **(30)**

A Do as directed (Any Three)

1 Python provides a standard GUI framework _____ which is used to develop fast and easy GUI applications

2 _____ is the error raised when a built-in function is inputted with invalid data type values

3 What is the correct keyword for creating an alias for a module?

4 If you do not know the number of keyword arguments that will be passed into your function, there is a prefix you can add in the function definition, which prefix?

B Do as directed (Any Three)

1 _____is the correct comparison operator to check if 5 is not equal to 10

2 _____ statement is used to stop a loop

3 _____ collection does not allow duplicate members

4 In Python, 'Hello', is the same as "Hello" True or False

C Do as directed (Any Three)

1 Which symbol is used to insert COMMENTS in Python code?

2 Multithreading in python is achieved through _____

3 A _____ is a function that modifies other functions

4 Python provides _____ modes to open files

D Do as directed (Any Three)

1 _____ modules include a function for accessing the filesystem

2 To remove the whitespaces and trailing spaces from the string, Python provides _____ built-in function **P.T.O.**

3 To remove leading characters from a string, we can use _____ function

dwHelpContext☻ ÅE¶☻

4 pvReserved☻xec☻☻pfnDeferredFree☻pELL☻♣scode☻☻ÉÉÉÿ[☻P

is a special literal used for defining a null variable

E Do as directed (Any Three)

1 _____ can be defined as a data which is given in a variable or constant

2 The _____ method works similarly to the constructors in Java

3 _____ in python can only contain elements of same data types

4 _____ is used to represent the instance of the class

SECTION-II

2. Answer the following: (Any Two) **(12)**

A Python is dynamically typed language, Explain

B What is the difference between .py and .pyc files?

C Discuss the Scope in Python?

3. Answer the following: (Any Two) **(12)**

A Discuss the modes for file read/write with example

B Elaborate the escape sequence with description

C Write to program which returns the count vowels in string

4. Answer the following: (Any One) **(06)**

A Discuss the regression techniques

B Discuss the applications of AI

END OF PAPER

Enrollment No._____

GANPAT UNIVERSITY
M.Sc. (CA&IT) SEM-II EXAMINATION (CBCS)
P22A2WDF: WEB DEVELOPMENT FRAMEWORK
April - June 2023

Time: 3 Hours] [Total Marks: 60

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

SECTION-I

1. Answer the following: (Any Five out of Seven) **(30)**

[A] What is framework? Explain MVC Framework in detail.
[B] Explain application architecture of Code-Igniter framework with figure.
[C] Discuss the various frameworks available in PHP.
[D] Explain Form validation in CodeIgniter.
[E] Discuss about application structure of Laravel.
[F] Explain the role of middleware in detail.
[G] What is Blade Templates? Describe various directives of Blade Templates.

SECTION-II

2. Answer the following. (Any Two) **(12)**

[A] Discuss the advantages and disadvantages of framework.
[B] Explain directory structure of CodeIgniter framework.
[C] What is Query Builder class in CodeIgniter? Describe any five function of Query Builder class.

3. Answer the following. (Any Two) **(12)**

[A] Differentiate library and helper. Describe session helper in codeigniter.
[B] Explain routing in Laravel.
[C] How does validation works in Laravel?

4. Answer the following. (Any One) **(06)**

[A] Discuss error handling in Laravel.
[B] Discuss about security in Laravel.

END OF PAPER

Seat No._____

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GANPAT UNIVERSITY

M.Sc. (CA & IT) SEM-II EXAMINATION
April - June 2023

P22A3FDS: FUNDAMENTALS OF DMM AND SEO

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks.
2. Each section should be written in a separate answer book.
3. Be precise and to the point in your answer.

SECTION-I

1. Answer the following: (Any Five) **(30)**

[A] Explain the different ways of Digital Marketing in current scenario.
[B] State the importance of Landing Page with example.
[C] Write steps to create Google My Business account.
[D] Discuss the Social Media Strategies for building a brand.
[E] Explain the Off-Page SEO factors in brief.
[F] Discuss the use of Robot.txt and Site map for blog.
[G] Write steps to submit the AdSense request for approval.

SECTION-II

2. Answer the following. (Any Two) **(12)**

[A] State the importance of Digital Marketing in 2023.
[B] Elaborate the features of an effective Email marketing campaign.
[C] Explain the Facebook Lead generation campaign in detail.

3. Answer the following. (Any Two) **(12)**

[A] Discuss the working mechanism of Search Engine in brief.
[B] State the importance of On-Page SEO factors in detail.
[C] Discuss the use of any one SEO Audit reporting tool with example.

4. Answer the following. (Any One) **(06)**

[A] Write a short note on Affiliate marketing,
[B] Discuss the Google Search and Display ads in detail.

END OF PAPER

Seat No._____

GANPAT UNIVERSITY

M.Sc. (CA & IT) SEM-II EXAMINATION
April - June 2023

P22A3FDS: FUNDAMENTALS OF DMM AND SEO

Time: 3 Hours] **[Total Marks: 60**

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END OF PAPER

GANPAT UNIVERSITY
M.Sc. (CA & IT) SEM-II EXAMINATION
April - June 2023
P22B4CDP2: CAREER DEVELOPMENT AND PLANNING-II

Time: 01:30 Hours] [Total Marks: 40

1. **Answer the following:** (40)

1. Every rational number is a-
   A) Whole number
   B) Real number
   C) Natural number
   D) None of the above
2. Digit 1 is occurring 136 times on writing all of the page numbers of a book. What will be the number of pages in the book?
   A) 194
   B) 195
   C) 200
   D) 295
3. To complete a piece of work, Samir takes 6 days and Tanvir takes 8 days alone respectively. Samir and Tanvir took Rs.2400 to do this work. When Amir joined them, the work was done in 3 days. What amount was paid to Amir?
   A) Rs. 300
   B) Rs. 400
   C) Rs. 800
   D) Rs. 500
4. Two numbers are respectively 20% and 50% more than a third number. The ratio of the two numbers is:
   A) 2:5
   B) 3:5
   C) 4:5
   D) 6:7
5. Odometer is to mileage as compass is to
   A) speed
   B) hiking
   C) needle
   D) direction
6. In a mixture 60 litres, the ratio of milk and water 2 : 1. If this ratio is to be 1 : 2, then the quantity of water to be further added is:
   A) 20 litres
   B) 30 litres
   C) 40 litres
   D) 60 litres

7. A vendor bought toffees at 6 for a rupee.How many for a rupee must he sell to gain 20%?
A) 3
B) 4
C) 5
D) 6

8. A, B and C can do a piece of work in 20, 30 and 60 days respectively. In how many days can A do the work if he is assisted by B and C on every third day?
A) 12 Days
B) 15 Days
C) 16 Days
D) 18 Days

9. A alone can do a piece of work in 6 days and B alone in 8 days. A and B undertook to do it for Rs. 3200. with the help of C, they completed the work in 3 days. How much is to be paid to C?
A) Rs. 380
B) Rs. 400
C) Rs. 600
D) Rs. 800

10. A person crosses a 600 m long street in 5 minutes. What is his speed in km per hour?
A) 3.6
B) 7.2
C) 8.4
D) 10

11. An aeroplane covers a certain distance at a speed of 240 kmph in 5 hours. To cover the same distance in 1 hours, it must travel at a speed of:
A) 300kmph
B) 360kmph
C) 600kmph
D) 720kmph

12. Look at this series: 7, 10, 8, 11, 9, 12, ... What number should come next?
A) 7
B) 10
C) 12
D) 13

13. Pointing to a photograph. Bajpai said, "He is the son of the only daughter of the father of my brother." How Bajpai is related to the man in the photograph?
A) Nephew
B) Brother
C) Father
D) Maternal Uncle

14. Look at this series: 21, 9, 21, 11, 21, 13, 21, ... What number should come next?
A) 14
B) 15
C) 21
D) 23

15. Tickets numbered 1 to 20 are mixed up and then a ticket is drawn at random. What is the probability that the ticket drawn has a number which is a multiple of 3 or 5?
A) 1/2
B) 2/5
C) 8/15
D) 9/20

16. A bag contains 2 red, 3 green and 2 blue balls. Two balls are drawn at random. What is the probability that none of the balls drawn is blue?
A) 10/21
B) 11/21
C) 2/7
D) 5/7

17. 120,99,80,63,48,?
A) 35
B) 38
C) 39
D) 40

18. In the series 2, 6, 18, 54, ... what will be the 8th term ?
A) 4370
B) 4374
C) 7443
D) 7434

19. 125,80,45,20,?
A) 5
B) 8
C) 10
D) 12

20. One morning Udai and Vishal were talking to each other face to face at a crossing. If Vishal's shadow was exactly to the left of Udai, which direction was Udai facing?
A) East
B) West
C) North
D) South

21. Y is in the East of X which is in the North of Z. If P is in the South of Z, then in which direction of Y, is P?
A) North
B) South
C) South-East
D) None of these

22. A F % 5 G H & 8 Y 6 3 N M * ! H 8 Which element is 3rd to the right of 5th from left?

A) &

B) 8

C) 6

D) 3

23. Y W @ 1 & C N 3 P L B 9 ↑ = D * E 2 £ M V ß 7 # 4 F G 5

If the numbers immediately preceding the symbols are attached to the value double their numerical value, then what will be the sum of the values of all sum numbers?

A) 22

B) 26

C) 36

D) 38

24. The total of the ages of Amar, Akbar and Anthony is 80 years. What was the total of their ages three years ago ?

A) 71 years

B) 72 years

C) 74 years

D) 77 years

25. Arrange the words in a meaningful sequence 1.Key 2.Door 3.lock 4.Room 5.Switch on

A) 5,1,2,4,3

B) 4,2,1,5,3

C) 1,3,2,4,5

D) 1,2,3,5,4

26. Arrange the words in meaningful sequence 1.Police 2.Punishment 3.Crime 4.Judge 5.Judgement

A) 3,1,2,4,5

B) 1,2,4,3,5

C) 5,4,3,2,1

D) 3,1,4,5,2

27. Arrange the words in meaningful sequence 1.Leaf 2.Fruit 3.Stem 4.Root 5.Flower

A) 3,4,5,1,2

B) 4,3,1,5,2

C) 4,1,3,5,2

D) 4,3,1,2,5

28. If A is the brother of B; B is the sister of C; and C is the father of D; how D is related to A?

A) Brother

B) Sister

C) Nephew

D) Cannot be determined

29. Introducing a boy, a girl said, "He is the son of the daughter of the father of my uncle." How is the boy related to the girl?
A) Brother
B) Nephew
C) Uncle
D) Son-in-law

30. How will you free the allocated memory?
A) remove(var-name);
B) free(var-name);
C) delete(var-name);
D) dalloc(var-name);

31. malloc() returns a NULL if it fails to allocated the requested memory.
A) True
B) False

32. Which is a valid keyword in java?
A) interface
B) string
C) float
D) unsigned

33. Java programs are
A) Faster than others
B) Platform independent
C) Not reusable
D) Not scalable

34. Java has its origin in
A) C Programming language
B) PERAL
C) COBOL
D) Oak programming language

35. Which data structure allows deleting data elements from front and inserting at rear?
A) Stacks
B) Queues
C) Deques
D) Binary search tree

36. Identify the data structure which allows deletions at both ends of the list but insertion at only one end.
A) Input-restricted deque
B) Output-restricted deque
C) Priority queues
D) None of above

37. Which of the following data structure is non-linear type?
A) Strings
B) Lists
C) Stacks
D) None of above

38. Which of the following data structure is linear-type?
A) Strings
B) Lists
C) Queues
D) All of above

39. When is the object created with a new keyword?
A) At run time
B) At Compile time
C) Depends on the code
D) None

40. Which of the following function displays information about PHP and its configuration?
A) php_info()
B) phpinfo()
C) info()
D) None of the above

END OF PAPER